Third week

NLP Internship

Learning is slow strength. Each step, even the hard ones, builds the mind you're becoming. Struggle is not failure—it's growth in progress. Keep going; tomorrow's clarity is built from today's persistence.

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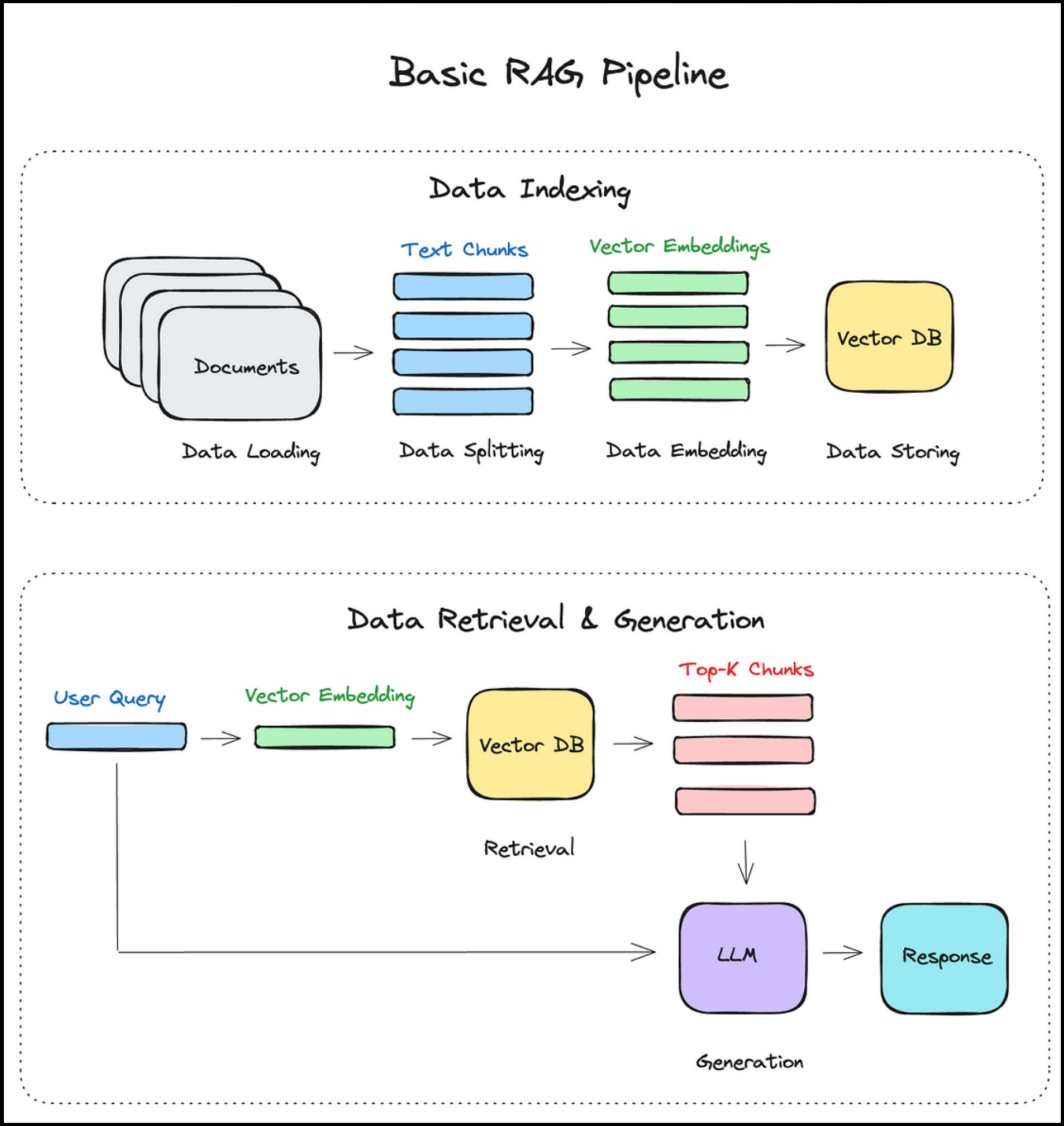
Basic RAG Pipeline
Data Indexing
Text Chunks
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Generation

First Watch these videos :

Part one : Introduction to RAG

[video 1] (20 min)

https://youtu.be/LIFyBPDkluI?si=xU8XpyJshSoST8dY

[video 2] (24 min)

https://youtu.be/s00dkGuK3rE?si=DJd77EciJqvvNgxm

[video 3] (31 min)

https://youtu.be/zxGWUJyB2V4?si=a_pdIvcLBomfe4io

[video 4] (25 min)

https://youtu.be/OGw023kOzR0?si=vVClyA9WByvdp8PJ

[video 5] (30 min)

https://youtu.be/sgRHmT896yY?si=mElV5cSd2Hzdgdr_

Part two : Vector database and FAISS

[video 1] (19 min)

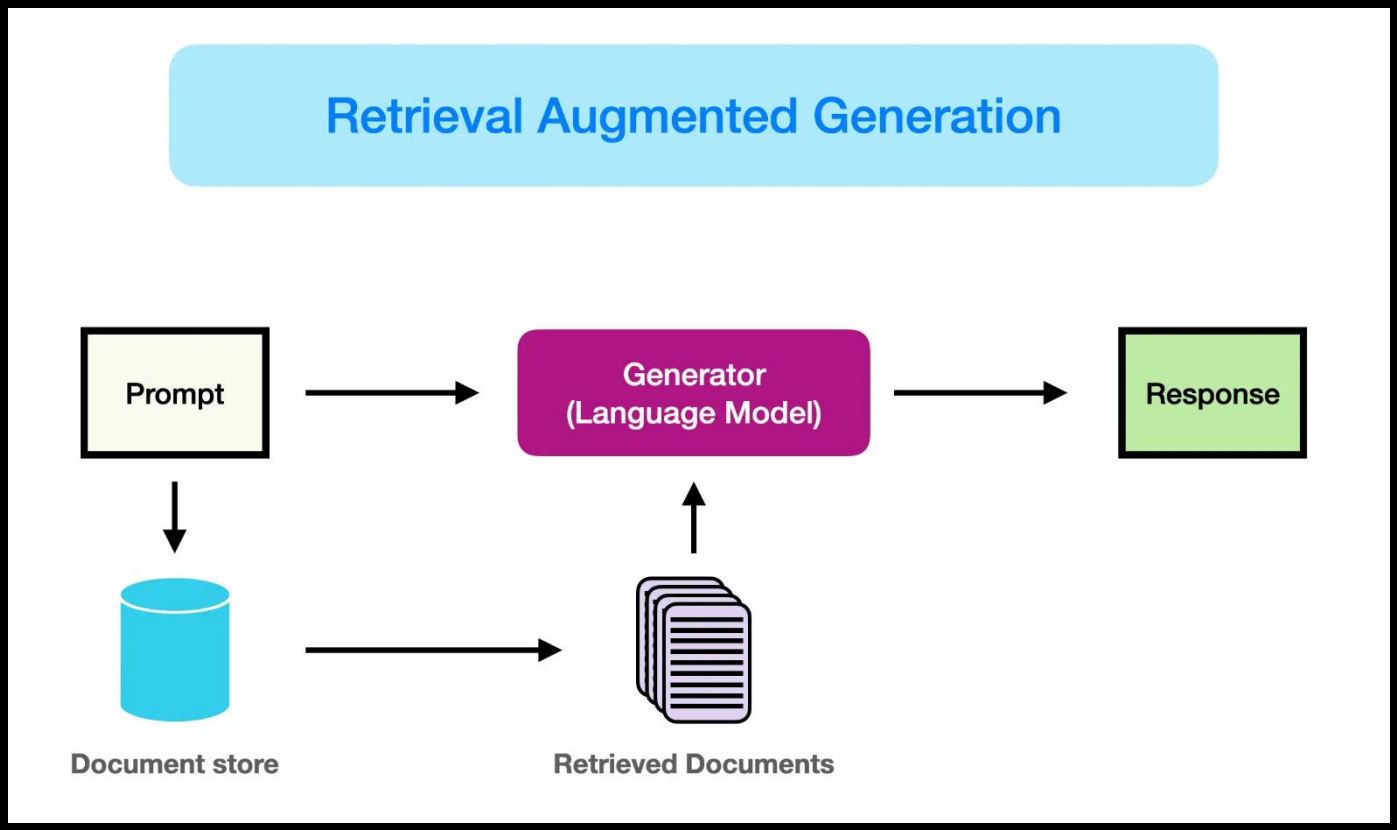
https://youtu.be/rp-IMDekA6I?si=Rb2SEmVQoDa00_HO

[video 2] (45 min)

https://youtu.be/reAmcocQyBA?si=8xgRrqeWzkVP1D8W

[video 3] (52 min)

https://youtu.be/VCnhiF88a9c?si=lYfEqUcrbxThjPVY



Part three : LangChain (Abu Bakr)

https://www.youtube.com/watch?v=A-sNuzZgY8g&list=PLvLvlVqNQGHDNUshQJBWWCIRGgC0PN7VL

Task

In this task, you are required to create a .txt file that contains detailed information about yourself. The file should include comprehensive definitions and descriptions, such as your background, interests, skills, education, projects, experiences, goals, and any other relevant personal or professional information. Make sure all important details are clearly written inside the file.

After creating the text file, you must build a Retrieval-Augmented Generation (RAG) system that can answer any question about you using the content of that file as the knowledge source.

The implementation must use LangChain to:

1. **Load and process the .txt file.**
2. **Split the text into appropriate chunks.**
3. **Generate embeddings.**
4. **Store the embeddings in a vector database.**
5. **Retrieve relevant chunks based on user queries.**

6. **Generate answers grounded strictly on the file's content.**

The final system should be able to respond accurately to questions about you by retrieving and using only the information stored in your text file.